



DATA ANALYSIS PROJECT REPORT

Project Title	Exploratory Data Analysis of PCOS Indicators
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Course / Section	Programming in Python-A
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1. Executive Summary

This project analyzes a PCOS medical dataset containing 2,000 patient records and 44 original variables. The main goal is to identify which clinical and physical factors are most associated with PCOS diagnosis among female patients using cleaned data, engineered features and visualizations. The analysis cleaned column names, converted numeric-looking text columns, filled missing numeric values with median values and engineered useful features such as total follicle count, LH/FSH ratio, BMI group, age group, symptom/lifestyle score and follicle level. The dataset contains 608 PCOS patients and 1,392 non-PCOS patients which has a PCOS rate of 30.40%. The strongest pattern appears in follicle count since PCOS patients have an average total follicle count of 20.62 compared with 9.09 for non-PCOS patients. PCOS patients also show higher average AMH, BMI, LH/FSH ratio and symptom/lifestyle score. The results suggest that follicle count, symptoms, AMH and BMI are important indicators in this dataset though the analysis cannot prove medical causation.

2. Introduction and Project Objectives

Polycystic Ovary Syndrome (PCOS) is a common health condition connected with hormonal, metabolic and reproductive indicators. This project uses exploratory data analysis to study how PCOS status relates to patient measurements, symptoms, lifestyle indicators and follicle counts. The aim is not to diagnose PCOS. The aim is to understand patterns inside the dataset and present them through clear tables and visualizations.

Item	Your Content
Dataset name	PCOS_extended_dataset.csv
Dataset source	https://www.kaggle.com/datasets/cm037divya/pcos-dataset
Main project goal	The main goal of this research is to explore and identify which clinical and physical factors are most associated with PCOS diagnosis among female patients using cleaned data, engineered features and visualizations.
Research Question 1	What is the distribution of PCOS and non-PCOS patients in the dataset?
Research Question 2	How does average total follicle count differ by PCOS status?
Research Question 3	Do BMI patterns differ between PCOS and non-PCOS patients?
Research Question 4	How does PCOS percentage change across BMI categories?
Research Question 5	What is the relationship between BMI and waist-hip ratio by PCOS status?
Research Question 6	Which selected numerical variables show stronger relationships with PCOS status?

3. Dataset Description

In the PCOS extended dataset each row represents one female patient. The dataset contains 2,000 patients and 44 columns. The columns has five categories of information such as demographics (age, weight, height, BMI, blood group), clinical and hormone measurements (FSH, LH, FSH/LH ratio, AMH, TSH, prolactin, progesterone, Vitamin D3, hemoglobin, random blood sugar, beta-HCG), physical measurements (pulse rate, respiratory rate, blood pressure, waist and hip measurements, waist-hip ratio), ultrasound findings (follicle count in left and right ovaries, average follicle size, endometrium thickness) and lifestyle and symptom indicators (weight gain, hair growth, skin darkening, hair loss, pimples, fast food consumption, regular exercise). The target variable is PCOS (Y/N). Here 1 indicates a PCOS diagnosis and 0 indicates none. There are 608 PCOS-positive and 1,392 PCOS-negative out of 2000 patients.

3.1 Basic Dataset Information

Property	Value
Number of rows	2000
Number of columns	44
Data types present	Numerical , Categorical , Text(Object)
Target or key variable(s)	pcos_y_n / pcos_status
Potential issues observed at first glance	Missing values, inconsistent column formatting, text-typed numeric columns, identifier columns, class imbalance.

3.2 Initial Observations

- The dataset is moderately imbalanced: 608 records are PCOS cases and 1,392 records are non-PCOS cases.
- Column names contained spaces, symbols, and inconsistent formatting, so they needed cleaning before analysis.
- Several variables are medical measurements while several others are binary symptom or lifestyle indicators.
- Extreme values were visible in some hormone and health-related columns so averages should be interpreted carefully.

4. Data Cleaning and Preparation

Issue Found	Action Taken	Reason / Justification	Code Reference
Messy column names with spaces and symbols	Standardized all column names to lowercase snake_case.	This made the code shorter, safer and easier to reuse.	<pre>data.columns = (data.columns .str.strip() .str.lower())</pre>

			<pre> .str.replace(r"^[a-z0-9]+", "_", regex=True) .str.strip("_")) </pre>
Duplicate rows	Checked and removed duplicate rows. No duplicates were found.	Duplicate patient records could distort counts and averages.	<pre> before_rows = data.shape[0] data = data.drop_duplicates() after_rows = data.shape[0] </pre>
Numeric-looking text columns	Converted ii_beta_hcg_miu_ml and amh_ng_ml to numeric values using pd.to_numeric.	These columns were needed for numerical summaries and correlations.	<pre> text_numeric_cols = ["ii_beta_hcg_miu_ml", "amh_ng_ml"] for col in text_numeric_cols: data[col] = pd.to_numeric(data[col], errors="coerce") </pre>
Missing numeric values	Filled missing numeric values with the median of each column.	Median is less affected by extreme values than mean.	<pre> numeric_cols = data.select_dtypes(include=["int64", "float64"]).columns for col in numeric_cols: if data[col].isnull().sum() > 0: data[col] = data[col].fillna(data[col].median()) </pre>
Identifier columns not useful for analysis	Dropped sl_no and patient_file_no after cleaning.	These columns identify records but do not explain PCOS related patterns.	<pre> data = data.drop(columns=["sl_no", "patient_file_no"], errors="ignore") </pre>

5. Feature Engineering

New Feature	How It Was Created	Why It Is Useful
pcos_status	Converted pcos_y_n into PCOS / No PCOS labels.	Makes charts and tables easier to read.
total_follicles	follicle_no_l + follicle_no_r	Combines left and right follicle count into one useful indicator.
lh_fsh_ratio	lh_miu_ml / fsh_miu_ml using np where when FSH > 0.	Helps compare hormone balance safely while avoiding division by zero.
age_group	Binned age into <=25, 26-30, 31-35, 36-40 and 40+.	Supports PCOS percentage comparison across age ranges.

bmi_group	Binned BMI into Underweight, Normal, Overweight and Obese.	Supports analysis of PCOS percentage by body weight category.
symptom_lifestyle_score	Sum of weight gain, hair growth, skin darkening, hair loss, pimples and fast food indicators.	Gives one simple score for symptom/lifestyle burden.
follicle_level	Binned total_follicles into Low, Medium and High.	Makes follicle-based PCOS comparison easier to explain.

6. Analysis Methodology

- Pandas was used to load the dataset, clean column names, remove duplicates, group records, calculate means, create summary tables and compare PCOS groups.
- NumPy was used for conditional feature creation for LH/FSH ratio calculation, z-score based outlier analysis and percentile calculations.
- Matplotlib was used to create six visualizations: patient-count bar chart, follicle comparison chart, BMI histogram, BMI-group PCOS percentage chart, BMI versus waist-hip scatter plot and correlation heatmap.

7. Analysis and Visualizations

7.1 Research Question 1

RQ1: What is the distribution of PCOS and non-PCOS patients in the dataset?

The PCOS status variable was counted after cleaning. The dataset contains 608 PCOS patients and 1392 non-PCOS patients. The PCOS rate is 30.40%.

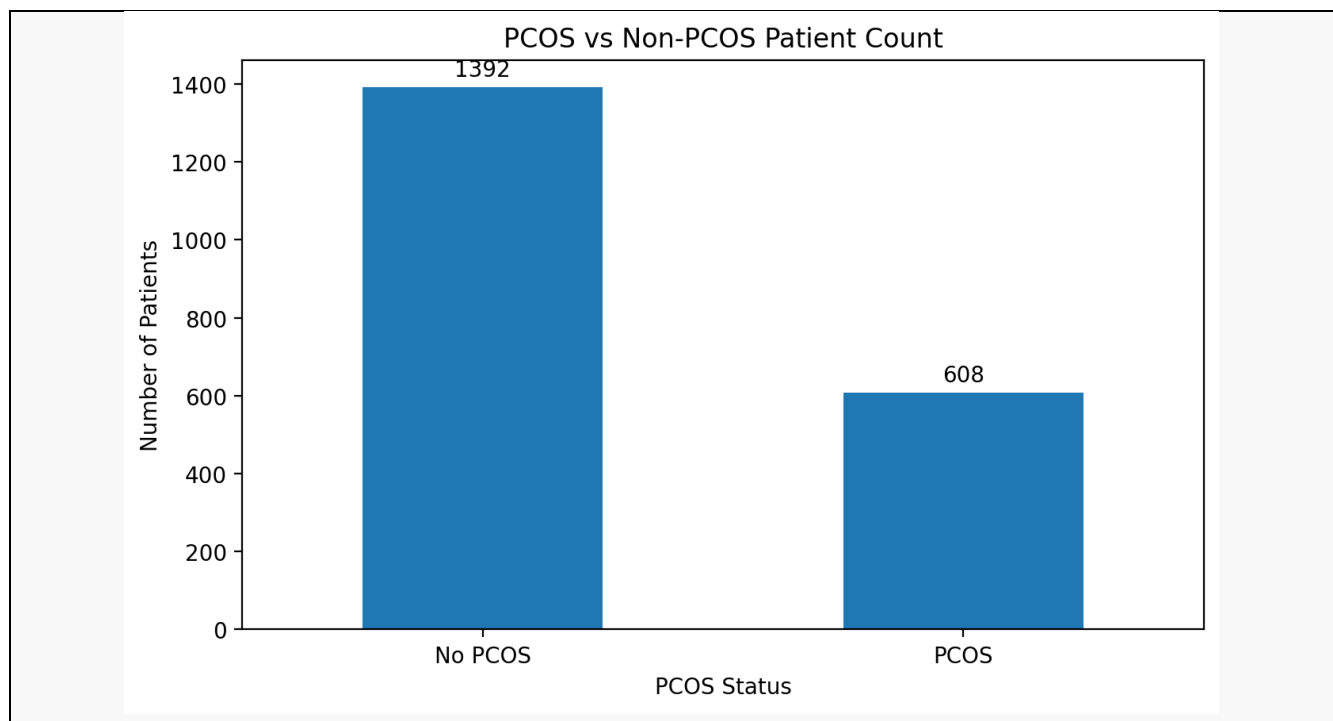


Figure 1. Distribution of PCOS and non-PCOS patient records.

Interpretation: The non-PCOS group is larger than the PCOS group. This means the dataset is imbalanced so percentage-based comparisons are more useful than raw counts alone.

7.2 Research Question 2

RQ2: How does average total follicle count differ by PCOS status?

Total follicle count was calculated by adding left and right follicle counts. Then the average total follicle count was compared between PCOS and non-PCOS groups.

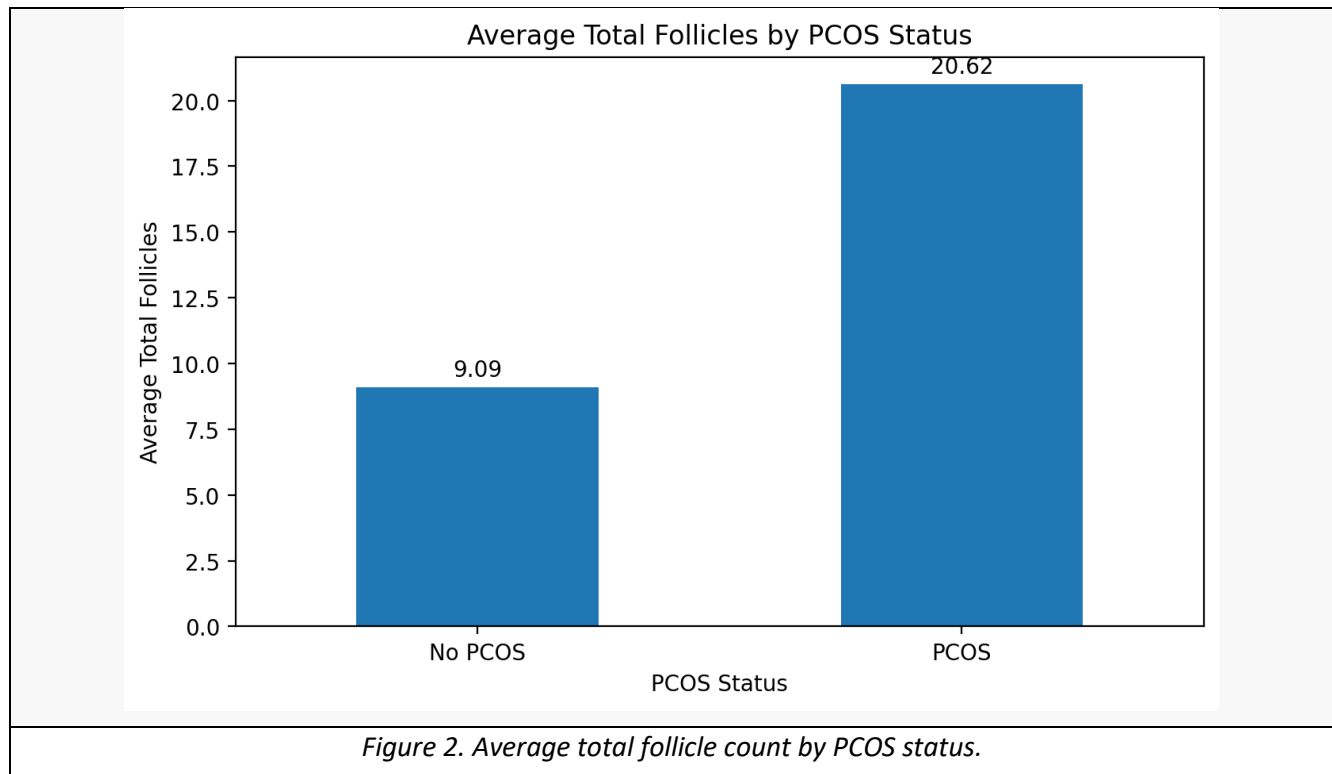


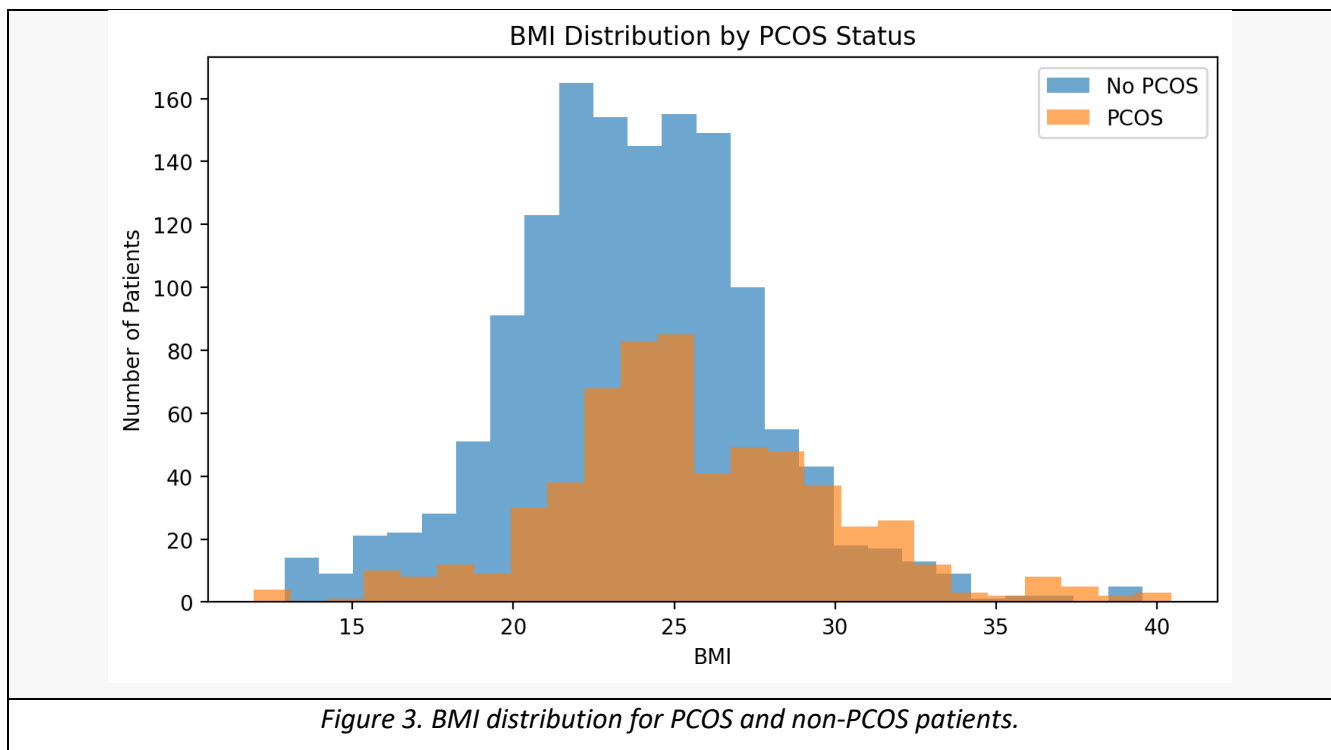
Figure 2. Average total follicle count by PCOS status.

Interpretation: PCOS patients have a much higher average total follicle count (20.62) than non-PCOS patients (9.09). This is the clearest pattern found in the project and strongly supports follicle count as an important PCOS-related variable in this dataset.

7.3 Research Question 3

RQ3: Do BMI patterns differ between PCOS and non-PCOS patients?

BMI values were separated by PCOS status and visualized through overlapping histograms. This helps compare the distribution of BMI rather than only comparing averages.



Interpretation: The PCOS group has a higher average BMI (25.56) than the non-PCOS group (23.73). The chart also shows overlap meaning BMI alone cannot separate PCOS and non-PCOS patients completely.

7.4 Research Question 4

RQ4: How does PCOS percentage change across BMI categories?

BMI was grouped into Underweight, Normal, Overweight and Obese categories. For each group the percentage of PCOS patients was calculated.

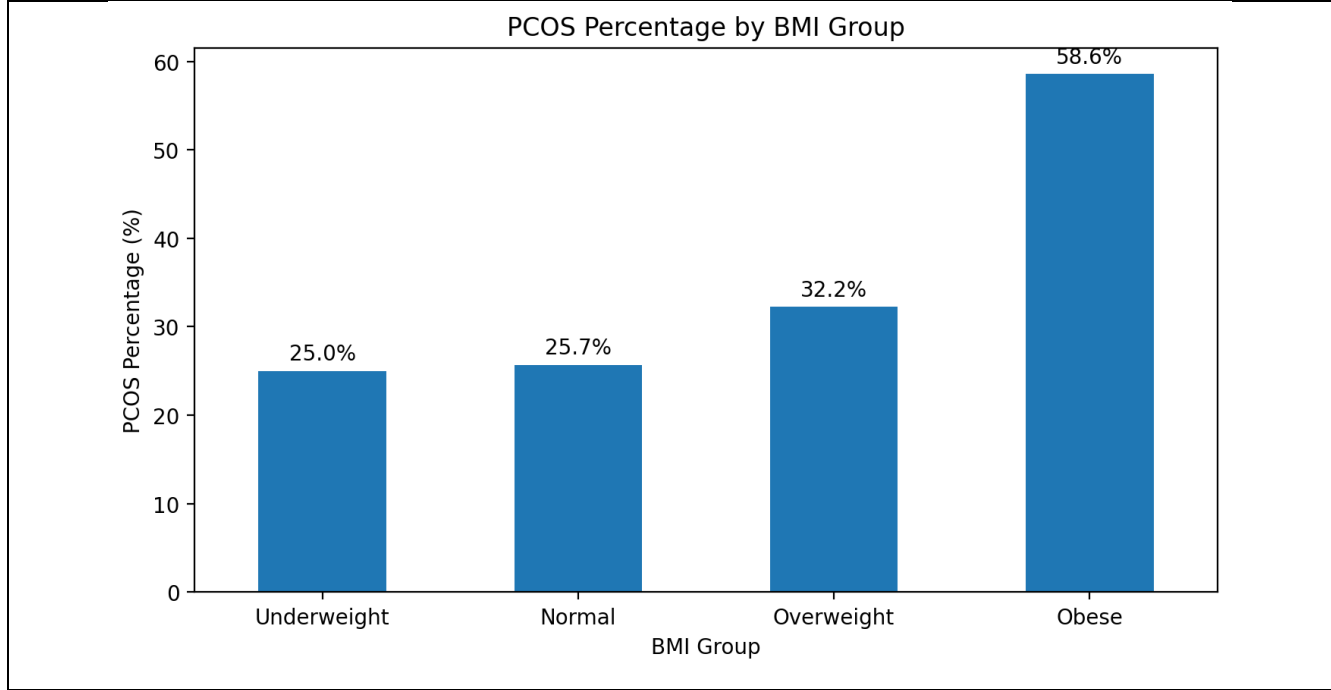


Figure 4. PCOS percentage across BMI groups.

Interpretation: PCOS percentage is lowest in the Underweight group (25.00%) and highest in the Obese group (58.60%). This suggests that higher BMI categories are linked with higher PCOS percentage in this dataset.

7.5 Research Question 5

RQ5: What is the relationship between BMI and waist-hip ratio by PCOS status?

A scatter plot was used to compare BMI and waist-hip ratio while separating PCOS and non-PCOS records. This shows spread and overlap between both groups.

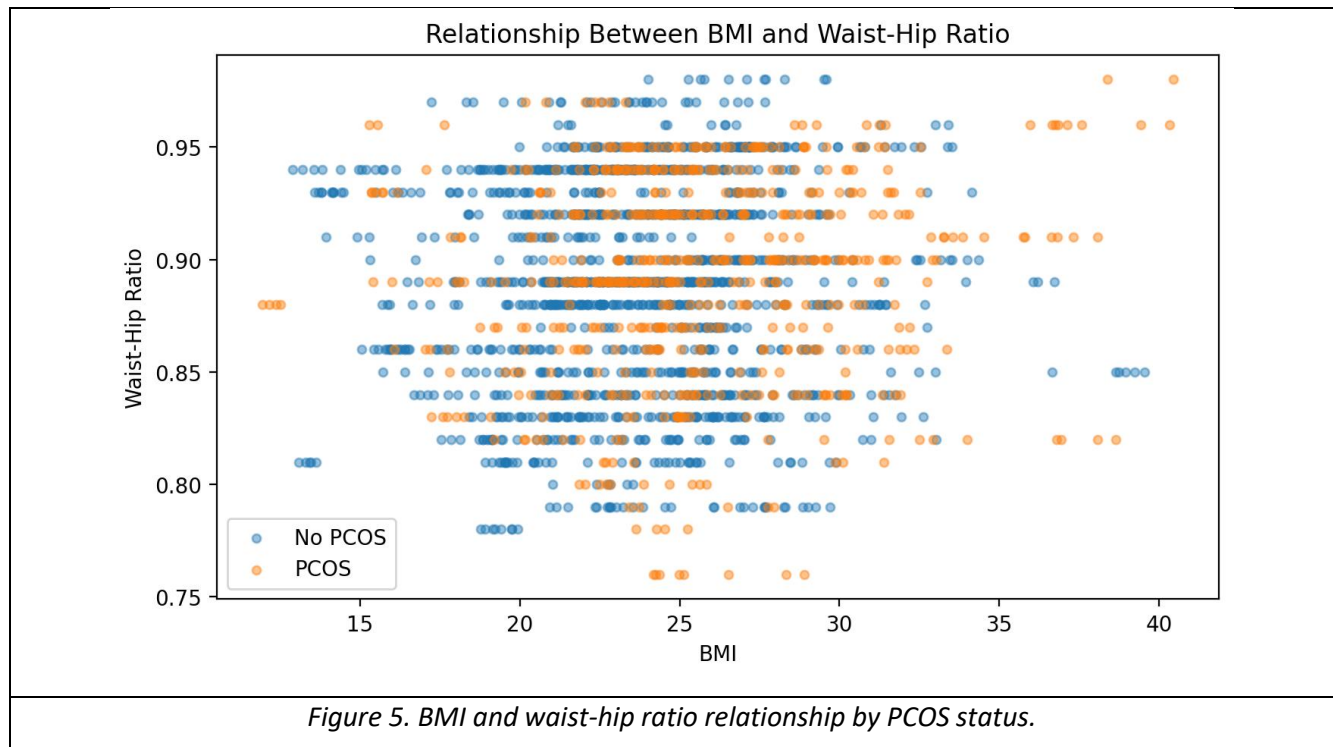


Figure 5. BMI and waist-hip ratio relationship by PCOS status.

Interpretation: Both groups overlap heavily in waist-hip ratio and their average waist-hip ratio is almost the same (0.89 for PCOS and 0.89 for non-PCOS). BMI differs more clearly than waist-hip ratio in this dataset.

7.6 Research Question 6

RQ6: Which selected numerical variables show stronger relationships with PCOS status?

A correlation matrix was created using selected numeric variables. Correlation with `pcos_y_n` was also calculated separately to identify variables more strongly related to PCOS status.

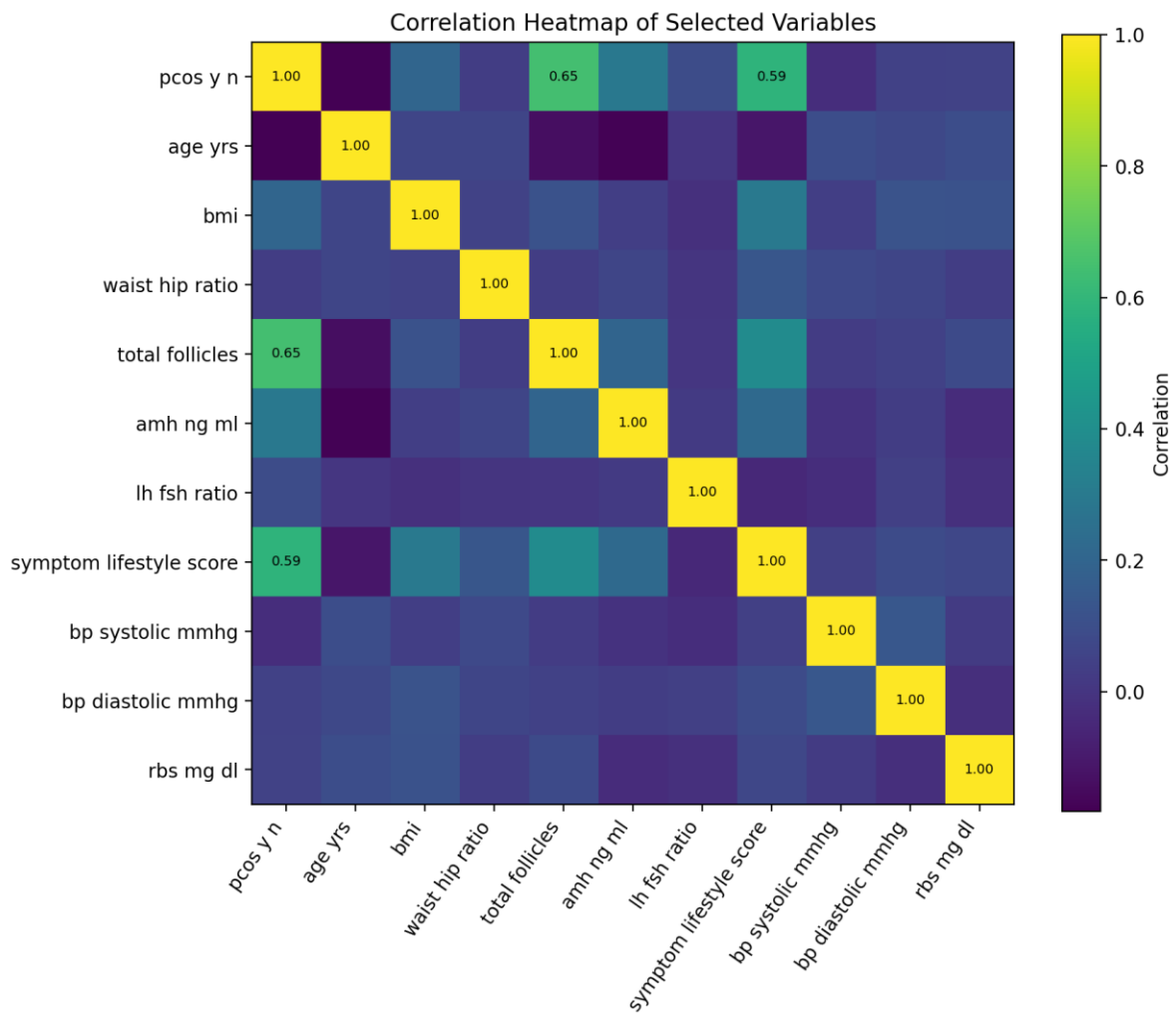


Figure 6. Correlation heatmap of selected numerical variables.

Interpretation: The strongest positive correlation with PCOS status is total_follicles (0.647) followed by symptom_lifestyle_score (0.587), AMH (0.290) and BMI (0.204). Age shows a negative correlation with PCOS status in this dataset.

8. NumPy-Based Custom Computation

Component	Description
Computation used	Z-score outlier calculation and percentile summary using NumPy
Purpose	Identify columns with unusual extreme values so the analysis can be interpreted carefully.
Formula or logic	$z = (\text{value} - \text{mean}) / \text{standard deviation}$. Values with absolute z-score greater than 3 were counted as outliers. Median, 95th percentile and maximum were also calculated with NumPy.
Result / takeaway	Several medical measurement columns contain outliers including AMH, total follicles, BMI, RBS, FSH, LH and Vitamin D3. This means averages should not be treated as perfect descriptions of every patient.

9. Key Findings

1. The dataset contains 608 PCOS patients and 1392 non-PCOS patients so PCOS cases make up 30.40% of the dataset.
2. Total follicle count is the strongest PCOS-related variable in this analysis. PCOS patients average 20.62 total follicles while non-PCOS patients average 9.09.
3. The symptom/lifestyle score is higher for PCOS patients (3.98) than non-PCOS patients (1.68).
4. PCOS percentage rises across BMI groups and reaches 58.60% in the Obese category.
5. AMH is higher in the PCOS group (8.87) than in the non-PCOS group (4.70).
6. The strongest positive correlation with PCOS status is total_follicles (0.647) followed by symptom_lifestyle_score (0.587).

10. Limitations

- This is an exploratory data analysis project not a medical diagnosis system.
- The dataset may not represent every population, region, age group or clinical setting equally.
- Some columns had missing values and some medical measurements had extreme values.
- Median filling is reasonable for basic analysis but it can still hide some missing-data patterns.
- Correlation does not prove causation. A strong relationship in this dataset does not mean one variable directly causes PCOS.
- No advanced statistical modeling or clinical validation was performed.

11. Conclusion

This project used Pandas, NumPy and Matplotlib to explore PCOS-related indicators in a 2,000-record dataset. The analysis found that total follicle count, symptom/lifestyle score, AMH and BMI have meaningful relationships with PCOS status. Among these total follicle count showed the strongest pattern. The PCOS group also had higher average BMI and higher symptom/lifestyle score than the non-PCOS group. At the same time the results must be interpreted carefully because the dataset contains missing values, outliers and observational relationships. Future work could use better clinical metadata, model validation and statistical tests to examine these patterns in more depth.

12. References / Dataset Source

Dataset: Kaggle — PCOS Dataset by cm037divya.

Source link: <https://www.kaggle.com/datasets/cm037divya/pcos-dataset>

Dataset file used: PCOS_extended_dataset.csv